

LINUX + GIT QUICK CHEAT SHEET

A one-page reference for basic terminal work, Git commands, and Git's three-tree architecture

1. BASIC LINUX COMMANDS

	Command	Sample usage	Purpose
	pwd	pwd	Shows the current folder path.
	ls	ls -la	Lists files, including hidden ones.
	cd	cd project	Moves into a directory.
	mkdir	mkdir assets	Creates a new folder.
	touch	touch index.html	Creates an empty file.
	cp	cp a.txt backup.txt	Copies a file or folder.
	mv	mv old.txt new.txt	Moves or renames an item.
	rm	rm notes.txt	Deletes a file; use carefully.
	cat	cat README.md	Displays file contents.
	grep	grep "error" log.txt	Searches text for a pattern.
	clear	clear	Clears the terminal screen.

2. BASIC GIT COMMANDS

Command	Sample usage	My description + common flags
git init	git init	Starts Git tracking in the current folder. Flags: -q quiet; --bare repository without a working folder.
git restore	git restore index.html	Brings a file back from the index or a commit. Flags: --staged unstage; --source choose a commit.
git status	git status	Shows changed, staged, and untracked files. Flags: -s short view; -b show branch.
git branch	git branch feature	Lists, creates, or deletes branches. Flags: -a all branches; -d delete; -m rename.
git add	git add .	Places selected changes into the staging area. Flags: -A all changes; -p choose chunks; -u tracked files.
git checkout / switch	git switch -c feature	Moves to another branch; switch is clearer for branches. Flags: -c create; checkout -b create branch.
git commit	git commit -m "Update home page"	Saves staged changes as a snapshot. Flags: -m message; -a stage tracked changes; --amend revise last commit.
git merge	git merge feature	Combines another branch into the current branch. Flags: --no-ff force merge commit; --abort cancel conflicted merge.
git log	git log --oneline --graph	Displays commit history. Flags: --oneline compact; -p patch details; --graph branch lines.
git tag	git tag v1.0	Labels a commit, often for releases. Flags: -a annotated; -m tag message; -l list.
git diff	git diff --staged	Compares file versions and shows edits. Flags: --staged index vs HEAD; --stat summary; --name-only filenames.
git clone	git clone https://github.com/user/repo.git	Copies a remote repository to the PC. Flags: -b chosen branch; --depth 1 shallow copy; --single-branch one branch.

3. GIT'S THREE-TREE ARCHITECTURE

1. Working Tree The actual project files on the computer. Edit, create, or delete files here. Typical flow: edit files → git add → git commit	2. Index / Staging Area The selected changes prepared for the next commit. Use git add . Check: git status shows what differs between the three areas.	3. HEAD / Repository The latest committed snapshot referenced by HEAD. Use git commit to save staged work. Compare: git diff = unstaged; git diff --staged = staged.
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References: Git Documentation — <https://git-scm.com/docs/git>; Git Cheat Sheet — <https://git-scm.com/cheat-sheet>; Git User Manual — <https://git-scm.com/docs/user-manual>; Linux man-pages — <https://man7.org/linux/man-pages/man1/git.1.html>
AI-use declaration: "The author used ChatGPT (GPT-5.6 Luna) to research, organize, and draft the cheat-sheet content and layout. The author reviewed and edited the content as needed and takes full responsibility for the final work."